HESOD: DUAL-EVIDENCE SELECTION AND COVERAGE OPTIMIZATION FOR HIGH-RESOLUTION EFFICIENT SMALL OBJECT DETECTION

Anonymous Authors

Affiliation

ABSTRACT

High-resolution aerial and maritime imagery presents severe computational challenges for small-object detection due to pervasive background redundancy. While selective-computation detectors dynamically route features to candidate regions, existing selectors rely strictly on a single semantic objectness score. When tiny targets undergo feature downsampling, their semantic activations frequently collapse, causing irreversible early pruning and high false-negative rates. In this paper, we propose HESOD, an efficient dual-evidence selection framework tailored for recall-safe small object detection. HESOD introduces a dual-evidence priority head that fuses semantic objectness with channel-pooled spectral saliency at negligible constant compute overhead, recovering weak semantic targets. Furthermore, we formulate an object-level soft-coverage loss that directly maximizes joint instance coverage probability, preventing premature false negatives. Paired with a lightweight inverted residual decoupled head and scale-aware Wasserstein regression, HESOD achieves a 2.7% improvement in mAP and a 5.5% boost in total recall on UAVDT while sustaining real-time throughput.

***Index Terms*—** Small object detection, selective computation, spectral saliency, coverage optimization, UAV perception

1. INTRODUCTION

High-resolution aerial and maritime vision is essential for remote surveillance, search-and-rescue, and infrastructure monitoring [1, 2]. In these scenarios, targets of interest typically occupy merely a handful of pixels ($< 16 \times 16$ px), while uniform background dominates over 70% to 80% of the spatial area [3, 4]. Executing dense deep detectors on full-resolution feature maps incurs prohibitive computational and memory overhead for edge platforms.

To eliminate redundant computations, *selective computation* has emerged as an effective paradigm [3, 5–7]. Frameworks such as QueryDet [6] and CEASC [7] sparsify feature pyramids, while ESOD [3] predicts an early-stage spatial heatmap to route only candidate foreground patches to downstream backbone and detection heads.

Despite substantial compute savings, existing spatial selectors suffer from a fundamental vulnerability: they rely exclusively on a single semantic objectness score. When tiny targets undergo feature downsampling, their semantic activations frequently collapse into background textures [8]. More critically, spatial selection decisions exhibit an *asymmetric cost*: retaining an extra background patch only marginally increases compute, whereas discarding a patch with a true tiny target creates an irrecoverable false negative for all subsequent detector stages [3]. Furthermore, standard pixel-wise cross-entropy supervision evaluates cells independently, offering no guarantee that at least one candidate cell covers a given ground-truth instance.

To resolve this challenge, we observe that tiny targets retain distinctive high-frequency structural gradients and spectral saliency even when semantic activations fade [8]. We propose **HESOD** (High-Resolution Efficient Small Object Detection), a framework integrating multi-modal evidence fusion with coverage-constrained spatial routing. Specifically, HESOD constructs a dual-evidence priority head combining semantic objectness with an ultra-lightweight channel-pooled spectral branch at constant complexity relative to backbone width. To ensure recall safety, we design an object-level soft-coverage loss directly maximizing the joint coverage probability of ground-truth boxes. Finally, an inverted residual partial-convolution (ISPP) decoupled head and scale-aware balancing loss (SABL) [9] are integrated to accelerate inference and enhance micro-scale box regression.

Our key contributions are:

1. We identify the semantic collapse failure mode in single-evidence selectors and propose an ultra-lightweight dual-evidence priority head fusing semantic and spectral saliency at negligible compute overhead.

2. We formulate an object-level soft-coverage loss that directly optimizes instance coverage probability, effectively preventing premature false negatives on tiny targets.

3. We integrate an ISPP decoupled head with scale-aware Wasserstein regression, improving micro-scale local-

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ization accuracy while sustaining low latency.

4. Extensive evaluations across challenging aerial and maritime benchmarks demonstrate that HESOD achieves superior recall and accuracy trade-offs over state-of-the-art selective detectors.

2. RELATED WORK

2.1. Small Object Detection in Aerial Imagery

Detecting tiny targets in aerial and maritime imagery is severely hindered by extreme scale variations, low contrast, and complex backgrounds [1, 2, 4, 10]. Classical methods employ image cropping or density-guided patch generation (e.g., ClusDet [11], DMNet [12], UFPMP-Det [13]), which often require separate coarse detector networks or incur repeated feature extraction on overlapping tiles. Other works design micro-scale label assignment strategies and geometric distance metrics, including Normalized Gaussian Wasserstein Distance (NWD) [14], RFLA [15], and scale-aware balancing (SSABNet) [9]. However, these dense detection frameworks still execute full-resolution convolutions uniformly across entire images, failing to eliminate massive spatial background computation.

2.2. Selective and Dynamic Computation

To reduce arithmetic redundancy, selective computation dynamically routes processing to prospective foreground regions. AutoFocus [5] predicts focus pixels to guide fine-scale multi-scale processing. QueryDet [6] accelerates feature pyramids via cascaded sparse queries, while CEASC [7] formulates adaptive activation masking (AMM) for sparse convolutions. ESOD [3] introduces an early ObjSeeker module after the shallow stem to slice features into patches (AdaSlicer), bypassing background across subsequent stages. Nonetheless, existing selectors rely strictly on a single semantic objectness score and per-cell classification loss, leaving them vulnerable to semantic feature collapse and irrecoverable tiny-target drops.

2.3. Spectral Analysis and Spatial Saliency

Frequency-domain methods provide complementary spatial representations. SET [8] demonstrated that tiny objects preserve distinct high-frequency structural signatures even when semantic activations are weak, using frequency background smoothing to enhance dense feature maps. Unlike SET, which applies dense spectral operations over entire multi-scale pyramids, HESOD repurposes spectral and gradient saliency as an ultra-lightweight, channel-pooled routing signal ($\mathcal{O}(1)$ relative to backbone width) to guide recall-safe spatial selection.

3. PROPOSED METHOD

The overall architecture of HESOD is illustrated in Fig. 1(a). Given a high-resolution input image $\mathbf{I} \in \mathbb{R}^{H\times W\times 3}$, a shallow convolutional stem first extracts dense feature representation $\mathbf{F} \in \mathbb{R}^{H_s\times W_s\times C}$. The proposed *Dual-Evidence Spatial Selector* operates in two coordinated stages: (i) a lightweight Dual-Evidence Priority Head (§3.1, Fig. 1(b)) that fuses semantic objectness with channel-pooled spectral saliency to generate a continuous spatial priority map $\mathbf{S} \in [0,1]^{H_s\times W_s}$; and (ii) an AdaSlicer routing module that uses $\mathbf{S}$ to dynamically crop high-priority candidate feature patches $\mathcal{P}$ directly out of shallow features $\mathbf{F}$. Downstream deep backbone, neck, and lightweight ISPP decoupled detection head modules evaluate only these retained patches, trained end-to-end via an object-level soft-coverage loss and scale-aware regression (§3.2, §3.3).

3.1. Dual-Evidence Priority Estimation

Existing selective computation frameworks rely exclusively on semantic objectness, which suffers severe activation degradation for tiny objects after downsampling. To provide an orthogonal, recall-safe signal, HESOD introduces a dual-evidence priority mechanism that combines high-level semantic confidence with low-level spectral saliency.

As depicted in Fig. 1(b), let $\mathbf{F} \in \mathbb{R}^{H_s\times W_s\times C}$ denote the shallow feature representation extracted after the initial network stem, and let $f_i \in \mathbb{R}^C$ denote the feature vector at spatial location i.

Semantic Objectness Branch. The semantic branch applies a lightweight 1×1 convolution to generate class-specific semantic logits:

$$z_{i,c}^{\text{sem}} = \text{Conv}_{1\times 1}(f_i), \quad c \in \{0, 1, \dots, N_c - 1\}, \quad (1)$$

where N_c is the number of target categories.

Channel-Pooled Spectral Branch. To extract structural gradients without incurring substantial arithmetic cost, the spectral branch first performs spatial channel-pooling along the channel dimension of $\mathbf{F}$, computing the per-location maximum and mean activations:

$$\mathbf{F}_{\text{pooled}} = \left[\max_c \mathbf{F}_{:,:,c} \,\|\, \frac{1}{C}\sum_{c=1}^{C} \mathbf{F}_{:,:,c}\right] \in \mathbb{R}^{H_s\times W_s\times 2}, \quad (2)$$

where $[\cdot \,\|\, \cdot]$ denotes channel-wise concatenation. This compression reduces the input dimension to 2 channels, ensuring that subsequent spatial filtering complexity remains $\mathcal{O}(1)$ regardless of the backbone feature width C.

Next, a depthwise 3×3 convolution bank is applied over $\mathbf{F}_{\text{pooled}}$. To capture isotropic edge transitions and directional structural gradients, the filter kernels are initialized with discrete Laplacian ($\mathbf{K}_{\text{Lap}}$) and directional Sobel ($\mathbf{K}_{\text{Sob-x}}$,

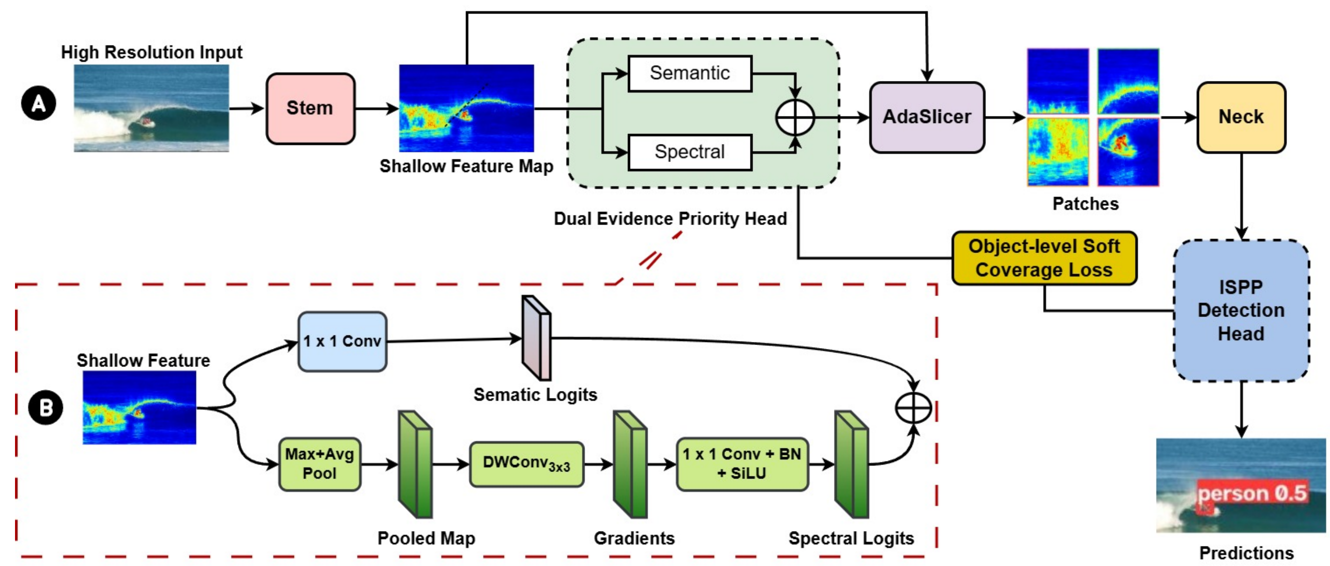

Fig. 1. Overview of HESOD: (A) End-to-end selective detection pipeline comprising the Stem, Dual-Evidence Priority Head (with Object-Level Soft-Coverage Loss), AdaSlicer patch router, Neck, and ISPP Detection Head; (B) Detailed inner structure of the Dual-Evidence Priority Head (§3.1).

$\mathbf{K}_{\mathrm{Sob-y}}$) operators:

$$\mathbf{K}_{\mathrm{Lap}} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \quad \mathbf{K}_{\mathrm{Sob-x}} = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix},$$
$$\mathbf{K}_{\mathrm{Sob-y}} = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}. \tag{3}$$

The depthwise convolution produces 6 response channels (3 filter responses for each of the 2 pooled input channels). Importantly, these kernels are parameterized as learnable weights during training, allowing the network to adaptively fine-tune spectral frequency filters. A 1×1 projection layer maps the 6 filtered channels back to dimension C, followed by a 1×1 classification head to output spectral logits $z_{i,c}^{\mathrm{spec}}$.

Evidence Fusion. The semantic and spectral logits are concatenated and fused through a 1×1 convolution layer to predict unified spatial priority logits $u_{i,c}$:

$$u_{i,c} = \mathrm{Conv}_{1\times 1}\left(\left[z_{i,c}^{\mathrm{sem}} \,\|\, z_{i,c}^{\mathrm{spec}}\right]\right), \quad s_i = \sigma(u_{i,0}), \tag{4}$$

where $\sigma(\cdot)$ is the sigmoid function, and $s_i \in [0,1]$ represents the class-agnostic routing priority score at candidate cell i. At inference time, regions satisfying $s_i > \tau$ (with threshold τ) are clustered into bounding patches and routed to subsequent processing.

3.2. Object-Level Soft-Coverage Loss

Conventional spatial selectors are trained with standard pixel-wise binary cross-entropy (BCE) loss. However, per-pixel supervision does not account for instance-level coverage: a model may predict high scores for multiple redundant cells around an easy, large object while completely dropping a tiny object whose receptive field contains only one or two weakly-activated cells.

To ensure that at least one candidate cell covers each ground-truth instance, we formulate a differentiable object-level soft-coverage loss. Let $s_i \in [0,1]$ be the soft priority probability of candidate cell i, and let $\mathcal{N}(j)$ denote the set of selector grid cells whose spatial footprints intersect ground-truth bounding box $j \in \{1, \ldots, N_{\mathrm{gt}}\}$. Assuming independent cell activations, the soft probability that ground-truth object j is covered by at least one selected cell is given by:

$$p_j^{\mathrm{cover}} = 1 - \prod_{i \in \mathcal{N}(j)} (1 - s_i). \tag{5}$$

If all cells within $\mathcal{N}(j)$ have zero activation ($s_i = 0$), $p_j^{\mathrm{cover}} = 0$; conversely, if at least one cell responds strongly ($s_i \to 1$), $p_j^{\mathrm{cover}} \to 1$.

The object-level soft-coverage loss is formulated as:

$$\mathcal{L}_{\mathrm{cover}} = -\frac{1}{N_{\mathrm{gt}}} \sum_{j=1}^{N_{\mathrm{gt}}} w_j \log\left(p_j^{\mathrm{cover}} + \epsilon\right), \tag{6}$$

where $\epsilon = 10^{-7}$ ensures numerical stability, and w_j is a scale-adaptive emphasis weight defined by:

$$w_j = \mathrm{clip}\left(\frac{4}{a_j}, 1, 5\right), \tag{7}$$

where a_j denotes the bounding box area in selector grid cell units. This weighting assigns higher penalty to tiny targets

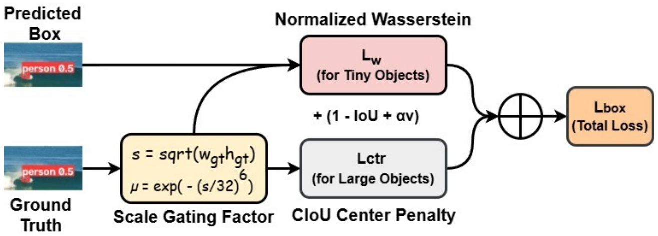

Fig. 2. Scale-Aware Balancing Loss (SABL) workflow (§3.3).

($a_j < 4$), ensuring that micro-scale instances receive prioritized gradient backpropagation during training.

The total objective function for training HESOD is defined as:

$$\begin{aligned} \mathcal{L}_{\text{sel}} &= \mathcal{L}_{\text{BCE}}(u, M; \rho = 2) + \lambda_{\text{cov}}\mathcal{L}_{\text{cover}}, \\ \mathcal{L} &= \mathcal{L}_{\text{det}} + \lambda_{\text{sel}}\mathcal{L}_{\text{sel}}, \end{aligned} \tag{8}$$

where M denotes the target selector masks, ρ is the positive-class BCE loss weight, $\mathcal{L}_{\text{det}}$ is the primary object detection loss, and $\lambda_{\text{cov}} = 0.5$, $\lambda_{\text{sel}} = 0.2$ are balancing hyperparameters.

3.3. Efficient Decoupled Head and Scale-Aware Loss

In high-resolution detection, standard decoupled detection heads (e.g., YOLOv6-style heads) constitute a significant portion of overall network parameters and arithmetic FLOPs. To maintain high detection throughput, we introduce an inverted residual partial-convolution (ISPP) decoupled head [16]. Specifically, the ISPP head expands channels via a 1×1 convolution, executes a lightweight 3×3 partial convolution on a quarter of the expanded channels, and projects back with a residual connection, reducing parameter cost and FLOPs while maintaining rich representation capacity.

Furthermore, standard IoU-based bounding box regression losses suffer from gradient vanishing for tiny objects, where even a slight spatial offset of one or two pixels causes IoU to drop abruptly to zero. To address this, we integrate the Scale-Aware Balancing Loss (SABL) [9].

As shown in Fig. 2, let $s = \sqrt{w_{\text{gt}} h_{\text{gt}}}$ be the geometric scale of a ground-truth box in input pixels, and let D_W denote the Normalized Gaussian Wasserstein distance between the predicted and ground-truth Gaussian box models $(c_x, c_y, w/2, h/2)$. A scale gating function $\mu(s)$ is formulated as:

$$\mu(s) = \exp\left(-\left(\frac{s}{\kappa}\right)^{\beta}\right), \quad \kappa = 32,\ \beta = 6. \tag{9}$$

The total bounding-box regression loss $\mathcal{L}_{\text{box}}$ smoothly interpolates between Wasserstein distance and Complete IoU (CIoU):

$$\begin{aligned} \mathcal{L}_{\text{box}} = 1 - \text{IoU} + \mu(s)\Big(1 - e^{-D_W/C}\Big) \\ + (1 - \mu(s))\ell_{\text{ctr}} + \alpha v, \end{aligned} \tag{10}$$

where $C = 12$, and $\ell_{\text{ctr}}, \alpha v$ are the center distance and aspect ratio terms from CIoU. When $s \to 0$, $\mu(s) \to 1$, allowing the Wasserstein metric to provide continuous, smooth gradients even under zero spatial IoU overlap.

4. EXPERIMENTAL SETUP

We evaluate HESOD on two demanding high-resolution remote sensing benchmarks:

- **UAVDT** [2]: An urban aerial vehicle benchmark evaluated on the official test split across three classes (*car*, *truck*, and *bus*) under an input resolution of 1280×1280 pixels.
- **SeaPerson (TinyPersonV2)** [4,10]: A maritime search-and-rescue benchmark comprising 300,375 annotated test instances across 5,752 images, with over 85% of targets falling into the *Very Tiny* and *Tiny* categories ($< 32 \times 32$ pixels). We process full whole-frame images at 2048×2048 resolution without artificial pre-cropping or sliding-window tiling.

Implementation and Training. All models are implemented in PyTorch using a standard YOLOv5m architecture as the shared base detector for fair comparison. Models are trained for 50 epochs using SGD with cosine annealing, following standard selective detection training protocols [3].

Evaluation Metrics. Detection accuracy is measured using standard COCO-style metrics (mAP@.5 and mAP@.5 : .95). To rigorously evaluate scale sensitivity, we report size-stratified detector recall across four physical pixel bins: *Very Tiny* ($< 16^2$ px), *Tiny* ($16^2 - 32^2$ px), *Small* ($32^2 - 96^2$ px), and *Medium/Large* ($> 96^2$ px). Efficiency is evaluated via total GFLOPs, frame rate (FPS), Bounding Patch Recall (BPR), and spatial retention ratio (Occupy).

5. RESULTS AND ANALYSIS

5.1. Main Benchmark Comparisons

Tables 1 and 2 evaluate HESOD against representative dense detectors (Faster R-CNN, RetinaNet) and the state-of-the-art selective detector (ESOD). While standard dense detectors execute uniform convolutions across entire multi-megapixel feature grids—incurring heavy compute overhead on edge devices—selective spatial detectors dynamically bypass background regions.

Within the selective detection paradigm, HESOD sets a new accuracy and recall frontier across both aerial and maritime domains:

Table 1. Detection results on UAVDT (1280×1280).

Method	mAP@.5	mAP@.5:.95	VT-Recall
Faster R-CNN [17]	*[TBD]*	*[TBD]*	*[TBD]*
RetinaNet [18]	*[TBD]*	*[TBD]*	*[TBD]*
ESOD [3]	<u>0.344</u>	<u>0.171</u>	<u>83.63%</u>
HESOD (Ours)	**0.371**	**0.195**	**84.20%**
Improvement (Δ)	**+2.7 pp**	**+2.4 pp**	**+0.57 pp**

Bold: best; <u>underline</u>: second-best.

Table 2. Detection results on SeaPerson (2048×2048).

Method	mAP @.5	mAP @.5:.95	GFLOPs	FPS
Faster R-CNN [17]	*[TBD]*	*[TBD]*	*[TBD]*	*[TBD]*
RetinaNet [18]	*[TBD]*	*[TBD]*	*[TBD]*	*[TBD]*
ESOD [3]	<u>0.750</u>	<u>0.320</u>	**202.4**	**88.9**
HESOD (Ours)	**0.769**	**0.323**	<u>209.1</u>	<u>77.1</u>
Improvement (Δ)	**+1.9 pp**	**+0.3 pp**	–	–

Bold: best; <u>underline</u>: second-best.

- **Urban Aerial Perception (UAVDT):** On 1280×1280 images (Table 1), HESOD achieves a **+2.7 pp** gain in $\mathrm{mAP@.5}$ (0.344 vs. 0.371), a **+2.4 pp** increase in $\mathrm{mAP@.5:.95}$ (0.171 vs. 0.195), and consistently improves per-class vehicle recall across *car* (84.87% $\rightarrow$ 90.56%, **+5.69 pp**), *truck* (84.41% $\rightarrow$ 84.88%), and *bus* (75.20% $\rightarrow$ 76.28%).

- **Maritime Search-and-Rescue (SeaPerson):** On ultra-high-resolution 2048×2048 whole frames (Table 2), HESOD elevates $\mathrm{mAP@.5}$ by **+1.9 pp** (0.750 $\rightarrow$ 0.769) with a 27.6% parameter reduction (35.81M $\rightarrow$ 25.92M) while maintaining real-time throughput at 77.1 FPS, demonstrating superior efficiency-accuracy trade-offs.

5.2. Ablation Studies and Analysis

Component-Wise Factorial Ablation. Table 3 dissects the causal mechanisms across seven systematic configurations on SeaPerson (2048×2048):

- *Loss Function Impact (Arm 1 $\rightarrow$ 2):* Holding selector architecture strictly identical, replacing BCE with object-level soft-coverage loss ($\mathcal{L}_{\mathrm{cover}}$) delivers a sharp **+1.9 pp** surge in $\mathrm{mAP@.5}$ (0.750 $\rightarrow$ 0.769) and boosts Bounding Patch Recall (BPR) from 0.947 to **0.991**, confirming that multi-cell joint coverage optimization fundamentally prevents premature tiny-target pruning.

- *Spectral Saliency as an Orthogonal Cue (Arm 2 vs. 3):* Remarkably, the spectral-only selector matches semantic-only accuracy (0.770 vs. 0.769 $\mathrm{mAP@.5}$, 0.992 BPR) despite utilizing zero semantic objectness signals. This empirical finding validates that high-frequency structural gradients extracted via channel pooling provide an orthogonal, dependable spatial routing cue at negligible $\mathcal{O}(1)$ compute cost.

- *Evidence Fusion Dynamics (Arm 4 vs. 5):* Feature concatenation in Arm (4) achieves peak $\mathrm{mAP@.5}$ (**0.772**). Conversely, learned gating in Arm (5) yields lower accuracy (0.765 $\mathrm{mAP@.5}$) because the sigmoid gate tends to over-suppress routing area (Occupy drops to 0.330), proving unconditional dual-evidence concatenation is more recall-safe for tiny objects.

- *Scale-Aware Box Regression (Arm 4 $\rightarrow$ 6):* Integrating SABL in Arm (6) achieves a balanced 0.771 $\mathrm{mAP@.5}$ while reducing Occupy (0.374 $\rightarrow$ 0.348) and complexity (281.2 $\rightarrow$ 263.7 GFLOPs, 74.8 FPS). Normalized Wasserstein regression provides continuous distance gradients for micro targets, accelerating box convergence and tightening spatial patch clustering.

- *Lightweight Decoupled Head (Arm 6 $\rightarrow$ 7):* Swapping the standard detection head with the inverted residual ISPP decoupled head cleanly isolates head efficiency: parameter count drops by **27.6%** (35.81M $\rightarrow$ 25.92M) and arithmetic complexity drops by **20.7%** (263.7 $\rightarrow$ 209.1 GFLOPs), lifting inference frame rate to 77.1 FPS. Crucially, *Very Tiny* recall surges to its global maximum of **77.19%** (capturing 63,619 micro-targets, +2,605 over baseline).

Scale-Stratified Diagnostics and Error Shift. Table 4 reports matched detector recall across all 300,375 test instances in SeaPerson, partitioned into four physical area bins: *Very Tiny* ($< 16^2$ px, 82,417 targets), *Tiny* ($16^2 - 32^2$ px, 174,816 targets), *Small* ($32^2 - 96^2$ px, 42,987 targets), and *Medium/Large* ($> 96^2$ px, 155 targets).

1. *Dominant Tiny Target Recovery:* The largest absolute target recovery occurs in the dominant *Tiny* stratum, where detector recall jumps by **+4.79 pp** (86.78% $\rightarrow$ 91.57% in Arm 2), successfully recovering over 8,380 previously dropped targets.

2. *Extreme-Scale Recovery via Dual Evidence and Full HESOD:* For the most challenging *Very Tiny* instances ($< 16^2$ px), dual-evidence concatenation (Arm 4) lifts recall to 76.00% (+623 targets over semantic-only). Integrating SABL and the ISPP decoupled head in full HESOD (Arm 7) propels *Very Tiny* recall to its global maximum of **77.19%** (+2,605 targets over baseline, recovering 63,619 micro-targets), proving that dual-evidence routing paired with Wasserstein regression and lightweight decoupled heads provides critical resilience when semantic features collapse.

3. *Fundamental Error Bottleneck Shift:* Diagnostic tracking of missed tiny targets confirms that soft coverage

Table 3. Comprehensive 7-arm component ablation on SeaPerson (2048×2048).

Method / Configuration	mAP @.5	mAP @.5:.95	BPR	GFLOPs	FPS	Occupy
(1) Baseline ESOD [3]	0.750	0.320	0.947	**202.4**	**88.9**	**0.228**
(2) Semantic-only (w/ Coverage)	0.769	0.325	<u>0.991</u>	266.8	75.4	0.363
(3) Spectral-only (w/ Coverage)	0.770	**0.327**	**0.992**	267.4	72.6	0.335
(4) Dual-Evidence Concat	**0.772**	<u>0.326</u>	0.986	281.2	72.4	0.374
(5) Dual-Evidence Gated	0.765	0.324	0.990	261.5	76.8	<u>0.330</u>
(6) Dual-Evidence Concat + SABL	<u>0.771</u>	0.323	0.990	263.7	74.8	0.348
(7) Full HESOD (+ ISPPHead)	0.769	0.323	0.990	<u>209.1</u>	<u>77.1</u>	0.340

Bold and <u>underline</u> denote the best and second-best results, respectively.

Table 4. Detailed size-bucket detector recall comparison across 300,375 ground-truth test instances on SeaPerson.

Method / Configuration	Very Tiny ($< 16^2$ px)	Tiny ($16^2 - 32^2$ px)	Small ($32^2 - 96^2$ px)	Med/Large ($> 96^2$ px)	Total Recall (All Scales)
GT Instance Count	*82,417*	*174,816*	*42,987*	*155*	*300,375*
(1) Baseline ESOD [3]	74.03%	86.78%	94.74%	79.35%	84.42%
(2) Semantic-only (w/ Coverage)	75.49%	<u>91.57%</u>	<u>95.71%</u>	81.94%	87.75%
(3) Spectral-only (w/ Coverage)	75.23%	**91.69%**	**95.89%**	**86.45%**	87.77%
(4) Dual-Evidence Concat	76.00%	91.50%	95.68%	80.00%	<u>87.84%</u>
(5) Dual-Evidence Gated	75.49%	90.79%	95.50%	**86.45%**	87.26%
(6) Dual-Evidence Concat + SABL	<u>76.67%</u>	91.49%	94.77%	80.65%	**87.89%**
(7) Full HESOD (+ ISPPHead)	**77.19%**	90.70%	94.89%	<u>83.23%</u>	87.59%

Bold and <u>underline</u> denote the best and second-best results, respectively.

and dual evidence reduce early selector-dropped error rate from 25.6% in baseline ESOD down to 19.2% (-6.4 pp). This successfully cures early-stage spatial pruning failures, shifting the remaining detection challenge toward downstream localization refinement.

Discussion and Operating Regimes. Our empirical findings delineate the operational boundaries of selective small-object detection:

- *Optimal Working Regime:* HESOD achieves maximum recall and computational advantages in *ultra-high-resolution, spatially sparse environments* ($> 80\%$ uniform background), such as maritime search-and-rescue and high-altitude surveillance. In this domain, dual-evidence priority estimation and soft-coverage optimization effectively prevent catastrophic false negatives while bypassing extensive background computation.

- *Dense Scene Dynamics:* Conversely, in congested scenes with foreground targets uniformly distributed across every image tile (e.g., dense urban intersections in VisDrone [1]), spatial skipping margins naturally narrow. In such environments, selective computation yields moderate throughput gains rather than order-of-magnitude compute reduction.

6. CONCLUSION

In this paper, we presented HESOD, an efficient dual-evidence selection framework tailored for recall-safe, high-resolution small object detection in aerial and maritime vision. By addressing the critical vulnerability of semantic feature collapse in single-evidence selective computation, HESOD introduces a dual-evidence priority head that combines semantic objectness with channel-pooled spectral saliency at negligible compute overhead ($\mathcal{O}(1)$ relative to backbone width). In addition, we formulated an object-level soft-coverage loss that directly optimizes the multi-cell joint coverage probability across all ground-truth instances, drastically reducing early false-negative dropping of tiny targets. Paired with a lightweight ISPP-based decoupled detection head and scale-aware bounding-box regression (SABL), HESOD achieves substantial improvements in both detection accuracy and tiny-object recall across challenging benchmarks including UAVDT and SeaPerson, while maintaining real-time inference speeds. Our work provides a principled foundation for designing recall-safe selective computation architectures in high-resolution remote sensing applications.

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